HOW IT WORKS

Finding the tumor and the fluid inside it

It finds the tumor’s bright, solid part and its dark, fluid-filled part, outlines both and suggests the type, using classic image processing only: no AI, no training.

What it uses

- Python, NumPy, SciPy
- scikit-image filters
- Brightness thresholds (Otsu)
- Shape rules and measurements

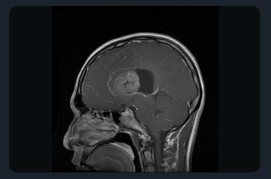
How it avoids AI

- No training data, no model
- Every decision is a written rule
- Same scan in, same answer out
- Every step can be shown

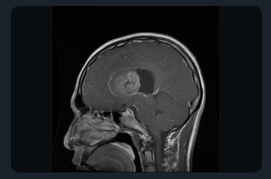
Finding the fluid

- Fluid is dark, like background
- So dark areas are checked too
- Must be round, thick, inside
- Mirrored pairs are normal

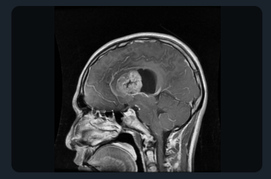
The steps



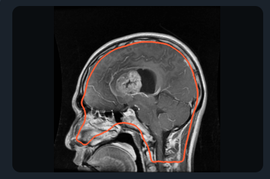
1 Load
Read the scan



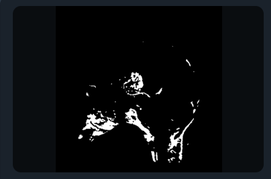
2 Denoise
Median filter



3 Enhance
CLAHE contrast



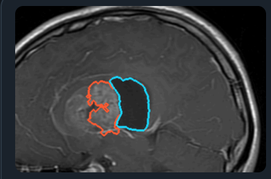
4 Remove skull
Threshold + erosion



5 Bright areas
Otsu thresholds



6 Fluid areas
New: dark, round, one-sided



7 Pick the tumor
Solid part + fluid

Glioma
Meningioma
Pituitary

8 Suggest type
Rules on place and look

Picture 7: orange = solid tumor, cyan = fluid inside it. Test patient 2935; 98% of the cyan is inside the doctor’s outline.